Name: _____ **Date:** _____

Balancing Chemical Equations

Balance the following chemical equations.

1. ____ KOH + ____ H_3PO_4 → ____ K_3PO_4 + ____ H_2O

2. ____ NH_3 + ____ O_2 → ____ NO + ____ H_2O

3. ____ Al + ____ O_2 → ____ Al_2O_3

4. ____ CaS + ____ H_2O → ____ $Ca(HS)_2$ + ____ $Ca(OH)_2$

5. ____ Au_2S_3 + ____ H_2 → ____ Au + ____ H_2S

6. ____ PCl_5 + ____ H_2O → ____ H_3PO_4 + ____ HCl

7. ____ Ba_3N_2 + ____ HF → ____ BaF_2 + ____ NH_3

8. ____ $CaCl_2$ + ____ Na_3PO_4 → ____ $Ca_3(PO_4)_2$ + ____ NaCl

9. ____ FeS + ____ O_2 → ____ Fe_2O_3 + ____ SO_2

10. ____ $HClO_4$ + ____ P_4O_{10} → ____ H_3PO_4 + ____ Cl_2O_7

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Answer key

Balance the following chemical equations.

1. 3 KOH + 1 H_3PO_4 → 1 K_3PO_4 + 3 H_2O

2. 4 NH_3 + 5 O_2 → 4 NO + 6 H_2O

3. 4 Al + 3 O_2 → 2 Al_2O_3

4. 2 CaS + 2 H_2O → 1 $Ca(HS)_2$ + 1 $Ca(OH)_2$

5. 1 Au_2S_3 + 3 H_2 → 2 Au + 3 H_2S

6. 1 PCl_5 + 4 H_2O → 1 H_3PO_4 + 5 HCl

7. 1 Ba_3N_2 + 6 HF → 3 BaF_2 + 2 NH_3

8. 3 $CaCl_2$ + 2 Na_3PO_4 → 1 $Ca_3(PO_4)_2$ + 6 NaCl

9. 4 FeS + 7 O_2 → 2 Fe_2O_3 + 4 SO_2

10. 12 $HClO_4$ + 1 P_4O_{10} → 4 H_3PO_4 + 6 Cl_2O_7

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